AWS Cloudfront

→ Content-Delivery-Network → Provides DDoS Protection.
→ Global ~~service~~ service.
→ Basically a caching service around the entire world.
→ Stores data at edge location.
→ Reduce latency.
→ 216 - Presence point of edge location.

AWS Global ~~Accele~~ Accelerator

→ Unicast IP → one server holds one IP
→ Anycast IP → all servers are given one IP only.

→ Basically Global Accelerator reduce latency by giving anycast IP to each and every resources you have in all the region.

→ Whenever a 'get' request is issueged, the server closest to the user, handles the request & return the output.

Best case → Use ~~both~~ three of them → CloudFront + Global Accelerator + Route 53

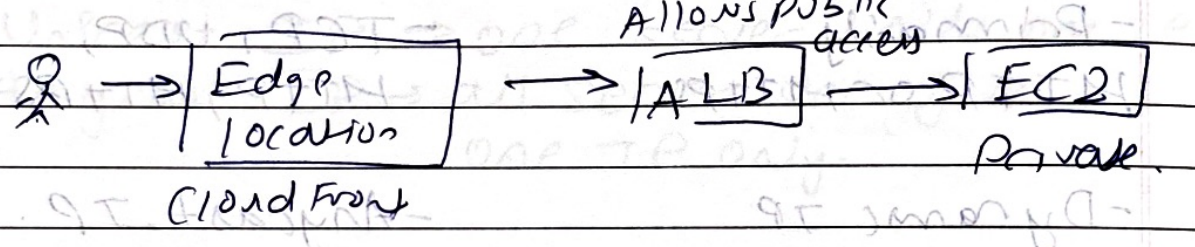
↑

Powerful & Globally distributed application.

Cloudfront vs	Global Accelerator
- CDN & caching	- Network availability & Performance.
- Best for cacheable content website, images, video	- Best for dynamic, non-cacheable traffic requiring consistent performance Eg- Gaming, IOT etc
- Primarily HTTP or HTTPS	- TCP, UDP, HTTP, HTTPS.
- Dynamic IP for each resource	- Anycast IP.
- Connect to various origins / ~~edge location~~ edge location	- Connects to AWS resources [ALB, NLBs, EC2 etc]
- Health checks only ~~on~~ on whether region is health or not. → combined with Route 53 to do health checks.	- Health checks directly to the resources.

CloudFront	vs	S3 Cross-Region replication
- Global Edge		- wherever you have replicated.
- Caching service TTL - 1 day after it is lost		- Stored permanently

→ To access private EC2 instances:



CloudFront for S3:

- For distributing files & caching them at edge.

- Secured using Origin Access Control [OAC]

CloudFront Geolocation Restriction:

As we have 216 location,

Allow List → Allow country which are in the list

Block List → Block country which are in the list.

→ Cloud Front pricing

As size ~~od~~ of data increases, pricing decreases
Eg:- 10 TB → 0.085 $
5 PB → 0.020 $

→ Cloudfront price class

All	Class 200	Class 100
All the region	- most region, but exclude most expensive region.	- Only the least expensive ones

Cloudfront - Cache Invalidation:-

→ See → suppose we have TTL set = 1 day so once visited, data is cached for one day

→ If we update & change data & now we want to update our website, how can we do it? TTL = 1 day so data is cached for 1 day & we can't change it.

→ Using cache invalidation, we can delete or update this cached data.